Testing by betting when the bets are real: an e-value audit of 84 pre-registered market-efficiency hypotheses

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Abstract

E-values are motivated by a gambler betting against a null hypothesis. We report an audit in which the gambler is not a metaphor: the null is a price posted by a bookmaker, the stake is a stake, and the wealth process *is* the e-process. The subject is a deployed UFC outcome forecaster and a pre-registered search for pockets where it beats the closing line — 84 hypotheses over 1,787 priced bouts, originally analysed with paired log-loss and Benjamini–Hochberg. Re-asked as bets, three things change. (i) The 84 slices are overlapping subsets of one pool, so BH's guarantee is not established; its one favourable discovery survives neither Benjamini–Yekutieli nor e-BH. (ii) A segment the fixed-n analysis reported as *failing to replicate* ($p = 0.30$) is, on the wealth scale, growing at the same rate in the replication window and simply 100 bouts short of $1/\alpha$ — optional continuation converts a refutation into a schedule. (iii) A post-hoc threshold rule, paid for by a mixture martingale, survives the search it actually ran ($e = 31.3$) and dies once the direction it could also have searched is priced in ($e = 15.6$). We report the ladder, not the winner.

1 A null hypothesis with money behind it

The literature on testing by betting [Shafer, 2021, Grünwald et al., 2024, Ramdas et al., 2023] introduces its central object through a gambler whose capital, by construction, cannot grow in expectation if the null is true. The gambler is usually a device. Here it is the situation itself.

We audit VERTEX, a deployed UFC bout-outcome model, against the closing line of a betting exchange. The substrate is a walk-forward out-of-fold pool: $4{,}141$ bouts scored between 2016 and 2026 by models refit at 32 quarterly origins, of which $1{,}787$ across 225 events carry an archived closing price. Two properties no simulation supplies:

The null is adversarial and external. H_0 is not an assumption of convenience — it is a number a counterparty posted with capital at risk, and a bettor who rejects it collects. Under H_0 the de-vigged closing probability q_t is the true conditional probability of the outcome $y_t \in \{0, 1\}$.

Predictability is enforced by the calendar. The model probability p_t comes from a fit that saw only bouts strictly before its origin, and q_t is posted before the fight. Neither has to be argued into being predictable.

The prior analysis of this pool [Prigodskii, 2026] declared 86 segment expressions — style matchups, durability, market microstructure, form, pedigree, division context — in a registry,[1] before scoring

[1] Pre-registration here means an author-controlled version history, not a third-party registration, and we do not claim more: the registry is a single source file whose commit (`8f9fcbd`, 2026-08-15T23:02:14+05:00)

any of them, each with a stated mechanism for why a good market would be wrong in that exact place. Two pairs proposed by different lenses are the same expression, leaving 84 distinct pre-registered hypotheses; three further slices were scored and marked post-hoc, and are excluded from the multiplicity accounting here as they were there. Those $84 + 3$ are the 87 in that report's title. It then measured each with a composition-adjusted paired log-loss contrast, cluster-robust by event, and charged multiplicity with BH. It also reported its own power: against a 3 percentage-point edge the paired test has 10–15% power, *falling to* 0–1% *once BH multiplicity is charged*, and would need $\sim$89,900 bouts. There are $1{,}787$. That report is the reason for this one: the instrument was known to be inadequate before its answers were read.

2 The wealth process

For a bout with de-vigged closing probability q and outcome y, and a predictable stake λ,

$$M \;=\; 1 + \lambda\,(y - q), \qquad \lambda \in \left(-\tfrac{1}{1-q},\, \tfrac{1}{q}\right), \qquad \mathbb{E}_{H_0}[M] = 1. \tag{1}$$

The product over a segment's bouts *in calendar order* is a non-negative test martingale; its terminal value is an e-value and, by Ville's inequality, its running maximum is anytime-valid. The growth-rate optimal stake under the alternative "the model's p is the truth" is available in closed form,

$$\lambda^\star \;=\; \frac{p - q}{q(1-q)}, \tag{2}$$

the model's edge deflated by the market's own variance. We report fractional Kelly at $\lambda = \frac{1}{2}\lambda^\star$, and, because that fraction is a free choice, also a predictable plug-in in the sense of Waudby-Smith and Ramdas [2024], which learns the multiplier from past bouts only (§7).

The margin is a supermartingale, not an excuse. Write $f = \lambda q$ for the fraction of wealth the stake commits to the backed side, and d for the decimal price the book actually offers on it; a fair book would offer $d = 1/q$. Settling at d instead gives $\mathbb{E}_{H_0}[M] = 1 - f(1 - q\,d) \le 1$, since $q\,d < 1$ whenever the book charges an overround. The deployable process is therefore a valid e-value that is uniformly conservative, short of the statistical one by exactly what the margin costs. On this pool the mean overround is 1.0482 and the shortfall is 0.00525 nats per bout: a measurement of how much statistical evidence the bookmaker's cut consumes.

Validation before use. Over $40{,}000$ replications under H_0, $\mathbb{E}[e] = 1.000 \pm 0.003$; $\Pr(\sup_t e_t \ge 1/\alpha)$ is 0.021, 0.058 and 0.142 at $\alpha = 0.05, 0.10, 0.20$; e-BH applied to 84 pure nulls rejects nothing; the confidence sequence of §6 covers the truth at *every* t in 98% of runs.

Where the model actually stands. Betting the model against the book over the whole priced pool gives $e = 3.6 \times 10^{-4}$ at fair odds and 3.1×10^{-8} at real ones. This is not a paper about beating a market, but about what can be honestly claimed inside a search that mostly failed.

3 The multiplicity charge BH was not licensed to make

The 84 slices are overlapping subsets of one pool of $1{,}787$ bouts, scored by one model against one book: a single upset moves dozens of them together. BH's FDR guarantee requires independence or PRDS [Benjamini and Yekutieli, 2001], and neither is established here. Nor is the dependence of the benign kind that would make PRDS plausible without an argument: across the $3{,}486$ segment pairs, membership correlations run from -0.49 to $+0.84$ with half of them negative, and 83 pairs share no bouts at all. The classical repair is Benjamini–Yekutieli, which charges $\sum_{i \le 84} 1/i = 5.01$. e-BH [Wang and Ramdas, 2022] is valid under *arbitrary* dependence and charges nothing.

On the discovery window the lab searched ($1{,}191$ bouts, the same 84 hypotheses):

procedure	guarantee holds?	rejects	
BH, paired statistic	not established	4	1 favourable, 3 against
Benjamini–Yekutieli ($\times 5.01$)	yes	0	
e-BH, wealth statistic	yes	0	cutoff $e \ge 1680$

precedes the first scored artifact (`6b5fbd9`, nine minutes later) in a public repository. The ordering and the contents are inspectable; the clock is the author's.

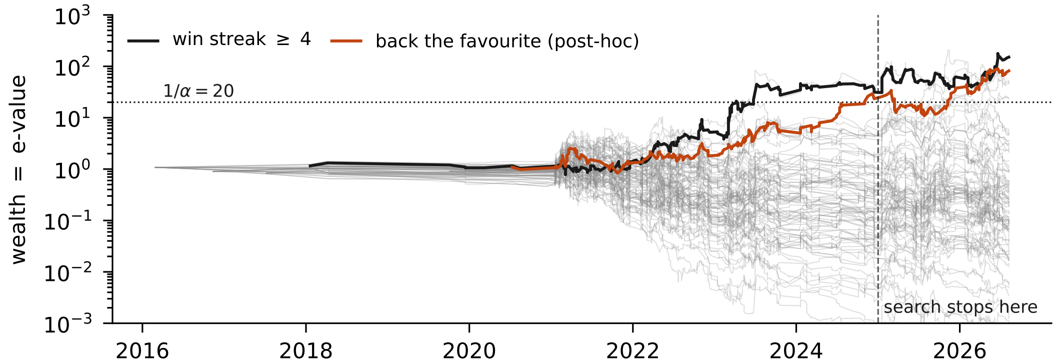

Figure 1: Wealth as a function of calendar time for every pre-registered segment (grey) and the two the analysis singled out (coloured). The dashed line is where the search stopped and the replication window began. A fixed-n analysis must start a new test to the right of it; a test martingale does not.

The lab's single favourable discovery — a fighter carrying a win streak of four or more, BH $q = 0.016$ — does not survive either procedure whose guarantee actually holds. The two valid procedures agree with each other and not with BH.

It would be wrong to read this as e-values buying less. The same segment's e-value on the full pool is 150.9, and for a *single* pre-specified hypothesis $e \geq 20$ is decisive evidence. What is expensive is not the machinery but the charge for having asked 84 questions of a pool this size — the same lesson the original analysis reached, now in a currency where the arithmetic is visible rather than delegated to a q-value.

4 "Failed to replicate" and "not yet" are different findings

The original analysis froze its discovery window, selected the streak segment, and ran a fresh fixed-n test on 2025–26. That test returned $p = 0.30$ and was reported, correctly under its own logic, as a failure to replicate. It is the only move a fixed-n design has: the second window buys a second test and nothing carries over.

On the wealth scale the same 110 bouts read differently. The segment's log-growth in the replication window is $+0.0143$ nats per bout against $+0.0174$ in the window that selected it: 82% of the discovery rate, the shrinkage one expects when a slice is chosen on the data that flatters it, but the same sign and the same order. The e-value accumulated there is 4.81 — real evidence, short of $1/\alpha = 20$. At that growth rate, crossing needs 210 bouts and 110 have arrived. The 210 is an extrapolation at a rate estimated on the same window, not a guarantee: the e-process is valid at whatever time it is stopped, but the date it crosses on is a forecast. The finding is not refuted. It is 100 bouts short, and this segment yields about 71 a year, so seventeen months and an e-process lets one simply keep multiplying rather than choose a stopping time in advance (Figure 1). We report the pre-split wealth ($e = 31.4$) but do not claim it: that is the window in which the segment was chosen.

The effect is stable across the five walk-forward seeds the pool carries (e-values 56.2 to 150.9, all above $1/\alpha$), which the q-value machinery had no natural place to put.

5 Pricing the freedom a post-hoc rule used

The largest effect in the original analysis is not a segment but a direction: where the model gives the book's favourite *more* than the book does, flat-staking that favourite returned $+11.5\%$ over 281 bets across 163 events. The analysis declared this post-hoc: the cut at $\mathrm{lean} \geq 0.05$ was chosen after seeing the sweep was positive from 0.02 to 0.22 and peaked at 0.13. A fixed-n framework has no principled penalty for that.

A mixture martingale has one, and it is a price fixed in advance rather than argued after: each threshold's wealth process is a martingale starting at 1, so any convex combination is too, and the search is paid for by the prior mass not placed on the winner. The maximum over thresholds is

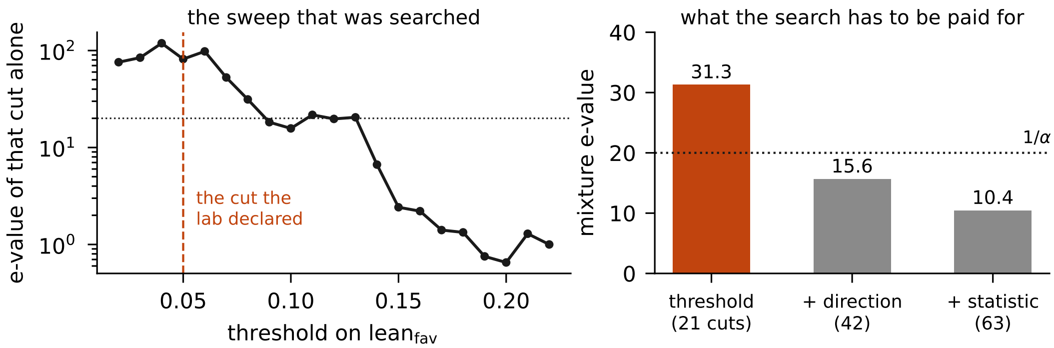

Figure 2: Left: the e-value of each threshold on the model's lean toward the book's favourite. The declared cut at 0.05 was chosen after this shape was visible. Right: a mixture martingale over progressively more of the freedom the search actually had. The finding survives paying for the threshold and does not survive paying for the direction.

118.99 and is not a valid e-value. The uniform mixture over the 21 cuts is $e = 31.3$, which clears $1/\alpha$. Widening the family to the 21 cuts the search could also have taken in the opposite direction gives $e = 15.6$, which does not. Adding the other disagreement statistic available on the same frame gives 10.4.

We regard the ladder, not any rung, as the result. It converts "this was post-hoc" from a caveat into a quantity: this finding can absorb the threshold search and cannot absorb the threshold search plus the choice of direction.

6 What the sequential instrument needs

Replacing the minimum detectable effect with a growth rate makes project planning a calculation rather than a verdict. Against a true 3 pp edge simulated on this pool's own price distribution, the half-Kelly bet grows at 0.00166 nats per bout and reaches $e = 1/\alpha$ in $\sim$1,800 bouts; the paired log-loss test needs $\sim$5,346 at 80% power. At 5 pp the figures are 648 and 1,902. The sequential test is roughly $3\times$ cheaper in fights, not because it is more powerful per observation, but because a paired contrast on a slice discards most of what each bout carries.

Anytime-validity is not free. For the directional rule, the cluster bootstrap gives a 95% interval on ROI of $[+2.2\%, +20.4\%]$ — valid at the one n at which the analysis stopped, which it chose after watching the number. A hedged betting confidence sequence [Waudby-Smith and Ramdas, 2024], valid at every n simultaneously, gives $[-5.3\%, +22.9\%]$ and does not exclude zero.

7 Limitations

The stake is a free choice. Fractional Kelly at $0.25/0.5/1.0$ gives $e = 26.0/150.9/115.3$ on the streak segment, and the predictable plug-in, which is not a choice, settles at $c = 0.70$ and $e = 63.1$. Every variant clears $1/\alpha$.

Boosting e-BH [Wang and Ramdas, 2022] would recover power but needs each e-value's null distribution, which a composite martingale null does not supply. Pre-specified family weights would be a better route, and were not specified.

Three contaminations are inherited from the pool and were sized there: an age correction whose shipping decision used replication-window data; hyperparameters tuned on a window the pool later scores (122 of $1{,}191$ discovery bouts); and a derived rating that is not fully point-in-time (21.9% of bouts, worth 0.0011 nats). Removing any of them moves the numbers unfavourably.

Finally, this is one sport, one book, and one model. The construction generalises to any forecaster audited against a market price; the numbers do not.

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